

Computer Science & IT

Database Management System



Relational Model & Normal Forms

Lecture No. 15



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Recap of Previous Lecture



Topic

Decomposition of relation



Topic

Multivalued dependency



Topics to be Covered



Topic

Properties of multivalued dependency



Topic

Fourth normal form (4NF)



Note:-

If a relation is in BCNF, then there will be 0% redundancy in that relation because of functional dependency, but such relation may still suffer from redundancy present because of Multi-valued dependency

Sid	Cid	Proj-No.
S ₁	C ₁	P ₁
S ₁	C ₁	P ₂
S ₁	C ₂	P ₁
S ₁	C ₂	P ₂
S ₂	C ₂	P ₂

Redundancy
Exist

it exist because of
Multi-valued dependency

$Sid \not\rightarrow Cid$
 $Sid \not\rightarrow Proj-No.$
 $Cid \not\rightarrow Sid$
 $Cid \not\rightarrow Proj-No.$
 $Proj-No. \not\rightarrow Sid$
 $Proj-No. \not\rightarrow Cid$
 $Sid, Cid \not\rightarrow Proj-No.$
 $Sid, Proj-No. \not\rightarrow Cid$
 $Cid, Proj-No. \not\rightarrow Sid$

i.e. ✓
 No non-trivial
 functional
 dependency
 exists in
 relation.
 ∴ Relation is
 at least in
 BCNF



Topic : Multivalued dependency

"X multivalued determines Y" is denoted by $X \twoheadrightarrow Y$

- ⊗ If there exist two or more independent attributes which are dependent on some other set of attributes then multi-valued dependency will exist in the relation

Formal definition of MVD:

Let R be the relation, and X & Y are some set of attributes from relation R ,

{ let us define

$$Z = \underbrace{\text{Attributes of } R - (X \cup Y)}_{\text{ie, } Z \text{ is a set of remaining attributes of } R}$$

If there exists at least 4 tuples $t_1, t_2, t_3, t_4 \in R$ such that,

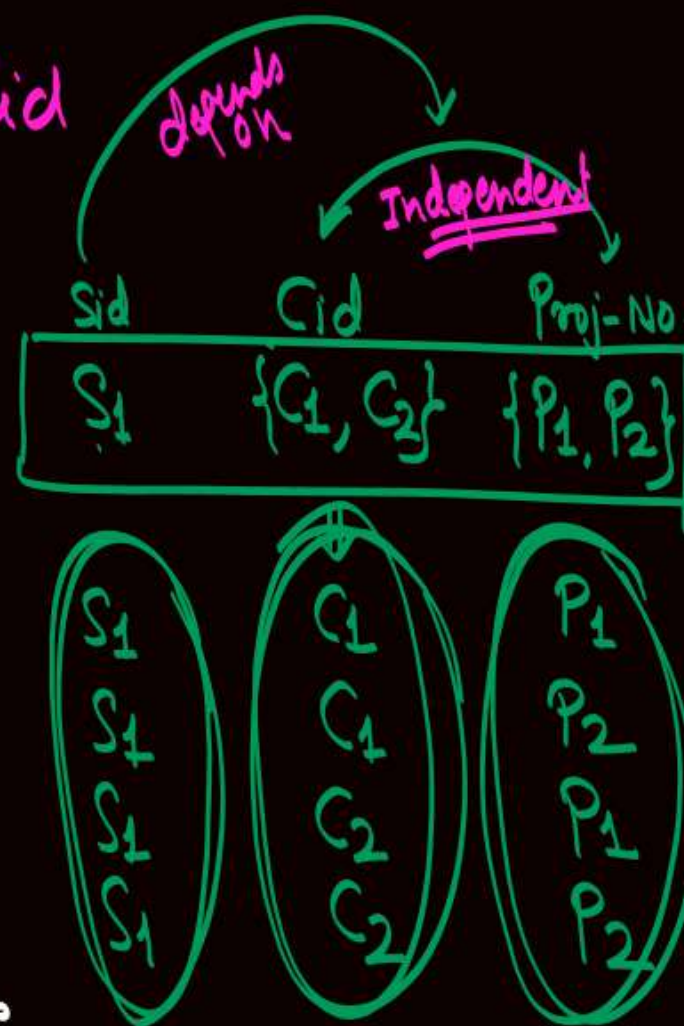
$$\left\{ \begin{array}{l} t_1.X = t_2.X = t_3.X = t_4.X \\ \text{and} \\ t_1.Y = t_2.Y \text{ and } t_3.Y = t_4.Y \\ \text{and} \\ t_1.Z = t_3.Z \text{ and } t_2.Z = t_4.Z \end{array} \right.$$

then Multi-valued dependency, $X \twoheadrightarrow Y$ exists in R .

then $Sid \twoheadrightarrow Proj-No.$ is also valid.



$Sid \twoheadrightarrow Cid$



	<u>X</u>	<u>Y</u>	<u>Z</u>
	Sid	Cid	Proj-No.
t1	S1	C1	P1
t3	S1	C1	P2
t2	S1	C2	P1
t4	S1	C2	P2
	S2	C2	P2

We can observe $\{ X=Sid, Y=Cid, Z=Proj-No \}$

$$t1.X = t2.X = t3.X = t4.X$$

And

$$t1.Y = t2.Y \text{ and } t3.Y = t4.Y$$

$$t1.Z = t3.Z \text{ and } t2.Z = t4.Z$$

$\Rightarrow \therefore X \twoheadrightarrow Y$
ie, $Sid \twoheadrightarrow Cid$ exists in Relation

* Another definition w.r.t. MVD :-

Whenever we swap the values of attributes of set Y
in two tuples (let t_1, t_2) which agree on the
values of the attributes of set X {i.e. $t_1.X = t_2.X$ },
and if the resulting tuples were already present
in the relation,

then $X \twoheadrightarrow Y$ in that relation
exist

	X Sid	Y Cid	Z Proj-No.
t ₁	S ₁	C₁	P ₁
t ₂	S ₁	C ₁	P ₂
t ₃	S ₁	C₂	P ₁
t ₄	S ₁	C ₂	P ₂
	S ₂	C ₂	P ₂

Resulting tuples
were already
present in
the relation

Does $Cid \twoheadrightarrow Sid$ exist in relation
 let $t_1 = (S_1, C_2, P_1)$
 & $t_2 = (S_2, C_2, P_2)$
 agree of the value of Cid
 ∴ let us swap the values of Sid
 ∴ Resulting tuples are

Not already
a member
 ∴ $Cid \twoheadrightarrow Sid$
 does not
hold true
in the
relation



Topic : Properties of Multivalued dependency

① Complementation Rule :- Let R be the relational schema and X & Y be some set of attributes over relation R ,

If $X \twoheadrightarrow Y$ exist in relation R .

then $X \twoheadrightarrow (\underbrace{\text{Attributes} - (X \cup Y)}_R)$ also exist in relation R

i.e. $(X \twoheadrightarrow Z)$ Let us call it Z , $Z = \underbrace{R - (X \cup Y)}_{\text{attribute wise}}$



Topic : Properties of Multivalued dependency

② Trivial MVD:-

Let $X \twoheadrightarrow Y$ exist in relation R ,

• $X \twoheadrightarrow Y$ is called a trivial MVD

if $X \supseteq Y$

$X \overset{\text{or}}{\cup} Y = R$ { i.e., Attributes of $X \cup$ Attributes of $Y =$ Attributes of R }



Topic : Properties of Multivalued dependency

③ Transitivity:-

If $X \twoheadrightarrow Y$ and $Y \twoheadrightarrow W$ exists in relation R

then $X \twoheadrightarrow (W - Y)$ will also exist in the relation.



Topic : Properties of Multivalued dependency

④ Augmentation :

If $x \twoheadrightarrow y$ exist in the relation R ,
and α, β are two set of attributes from
relation R , such that $\alpha \supseteq \beta$,

then $x\alpha \twoheadrightarrow y\beta$ will also hold true in relation R .



Topic : Properties of Multivalued dependency

⑤ Replication:-

- * Every functional dependency is also a M.V.D.
 { Every MVD need not be a functional dependency }

i.e., if $X \rightarrow Y$ exist in R , then $X \twoheadrightarrow Y$ will also exist in R

& if $X \twoheadrightarrow Y$ exist in R , then $X \rightarrow Y$ need not exist in R



Topic : Properties of Multivalued dependency

⑥ Splitting

Multivalued dependencies are not allowed to split.

i.e. if $X \twoheadrightarrow YW$ exist in relation, then, $\begin{cases} X \twoheadrightarrow Y \text{ and } X \twoheadrightarrow W \\ \text{need not hold true.} \end{cases}$

But FDs are allowed to split.
i.e. if $X \rightarrow YW$, then $\begin{matrix} X \rightarrow Y \\ X \rightarrow W \end{matrix}$



Topic : Fourth normal form (4NF)

For a relation R to be in 4NF.

① Relation R must be in BCNF,

and ② Every non-trivial MVD $X \twoheadrightarrow y$.

must be with X as a super key of the relation

→ Super keys and Candidate keys
are always determined
w.r.t functional dependencies.

Because, FDs are key dependencies
whereas, MVDs are data dependency.

{ i.e., In every
non-trivial FD $x \rightarrow y$,
x must be a
S.K. of relation.

Sid	Cid	Proj-No.
S ₁	C ₁	P ₁
S ₁	C ₁	P ₂
S ₁	C ₂	P ₁
S ₁	C ₂	P ₂
S ₂	C ₂	P ₂

Following MVD
exists in Relation

$$(i) \text{ Sid } \twoheadrightarrow \text{ Cid}$$

Not a S.K.
neither $\text{Sid} \supseteq \text{Cid}$
nor $\{ \text{Sid} \cup \text{Cid} \} = \text{All attributes}$

∴ A non-trivial MVD $X \twoheadrightarrow Y$
in which X is not a S.K.

$$(ii) \text{ Sid } \twoheadrightarrow \text{ Proj-No.}$$

Not a S.K.

neither $\text{Sid} \supseteq \text{Proj-No.}$
nor $\{ \text{Sid} \cup \text{Proj-No.} \} = \text{All attributes}$

∴ A non-trivial MVD $X \twoheadrightarrow Y$
in which X is not a S.K.

∴ Relation
is in
BCNF,
but not in
4NF

No non-trivial functional
dependencies exists in
the relation,

∴ Relation R is in BCNF.

And hence C.K. of Relⁿ = (Sid, Cid, Proj-No.)

→ If the relation is in BCNF, but not in 4NF,

then for 4NF decompose the relation w.r.t.

Every non-trivial MVD $X \twoheadrightarrow Y$, in which X is not a S.K.

Sid	Cid	Proj-No.
S ₁	C ₁	P ₁
S ₁	C ₁	P ₂
S ₁	C ₂	P ₁
S ₁	C ₂	P ₂
S ₂	C ₂	P ₂

Following MVD exists in Relation

(i) $Sid \twoheadrightarrow Cid$

Not a s.k.
neither $Sid \supseteq Cid$
nor $\{Sid \cup Cid\} = \text{All attributes}$

(ii) $Sid \twoheadrightarrow Proj_No.$

Not a s.k.

neither $Sid \supseteq Cid$
nor $\{Sid \cup Cid\} = \text{All attributes}$

Not in 4NF

For 4NF,
decompose
w.r.t.
non-trivial
MVDs

$Sid \twoheadrightarrow Cid$

Sid	Cid
S ₁	C ₁
S ₁	C ₂
S ₂	C ₂

$Sid \twoheadrightarrow Proj_No.$

Sid	Proj-No.
S ₁	P ₁
S ₁	P ₂
S ₂	P ₂

No. non-trivial FD
∴ BCNF.

The only MVD that exists in the relation is $Sid \twoheadrightarrow Cid$ and it is also trivial
∴ 4NF

No. non-trivial FD
∴ BCNF.

The only MVD that exists in the relation is $Sid \twoheadrightarrow Proj_No.$ and it is also trivial
∴ 4NF

No non-trivial functional dependencies exist in the relation,

∴ Relation R is in BCNF.

And hence C.k. of Relⁿ = (Sid, Cid, Proj-No.)



2 mins Summary



Topic

Properties of multivalued dependency ✓



Topic

Fourth normal form (4NF) ✓



THANK - YOU